

JYOTHSNA A

Data Engineer

+1(667)-910-3952 abbojujyo@gmail.com [linkedin.com/in/abboju-jyothsna](https://www.linkedin.com/in/abboju-jyothsna)

Professional Summary

- Over 5+ years of experience in data engineering, specializing in building scalable, resilient data systems with a focus on optimized Python programming for data processing, automation, and data-driven applications.
- Proficient in designing and managing data pipelines using Apache Spark, Beam, and Airflow, supporting both batch and streaming data, with advanced optimization techniques to ensure efficient, reliable data flow.
- Strong expertise in leveraging AWS services like S3, EMR, Redshift, and Glue for building robust data architectures, implementing ETL processes, and ensuring data quality and scalability across large datasets.
- Skilled in SQL and NoSQL databases, including Microsoft SQL Server, MongoDB, and DynamoDB, for effective data management, performance tuning, and ensuring data integrity and reliability in high-volume environments.
- Experienced in DevOps practices with Docker, Jenkins, and Terraform for CI/CD automation and infrastructure management, along with containerization to streamline deployments and enhance system performance.
- Hands-on experience with Apache Kafka for real-time data streaming, monitoring with Splunk, and machine learning integration using Spark MLlib, effectively supporting data-driven decision-making and ensuring high availability and fault tolerance.

Technical Skills

Languages: Python, Java, Scala, C++, NodeJS, .Net R, PL/SQL

Big Data Technologies: Apache Kafka, Apache Spark, Apache Beam, Hadoop YARN, Apache Hive

Cloud Services: AWS (S3, EMR, Redshift, Athena, Kinesis, RDS, Glue, Lambda, CloudWatch, IAM)

Database Technologies: Relational Databases (Microsoft SQL Server, PostgreSQL), Non-Relational Databases (MongoDB, DynamoDB, Cassandra), AWS Redshift, Snowflake

DevOps and Orchestration Tools: Docker, Git, Jenkins, Terraform, Apache Airflow

Business Intelligence & Visualization: Amazon Quick Sight, Tableau, Power BI

ETL Tools: Informatica PowerCenter, SSIS, Talend,

Data Pipeline and Analytics Tools: Apache Beam, Databricks, Splunk, Kibana

Methodologies: Agile, Business Analytics, Documentation, CI/CD

Machine Learning: TensorFlow

Experience

Citi

August 2023 – Present

Data Engineer

New York, US

- Architected and implemented efficient data pipelines using Apache Beam and Apache Spark, optimizing workflows for 30% improved performance and scalability.
- Developed and optimized ETL processes using AWS Glue and Apache Kafka, ensuring seamless data integration, transformation, and real-time streaming across diverse applications.
- Automated complex data workflows with Apache Airflow, enhancing operational efficiency, reliability, and overall pipeline orchestration.
- Built and optimized data pipelines on Databricks, leveraging Apache Spark for efficient data processing, transformation, and analysis of large datasets.
- Implemented scalable and cost-effective data warehousing solutions on Snowflake, enabling robust data storage, retrieval, and advanced analytics.
- Managed and optimized data lakes using AWS S3, Redshift, and EMR, ensuring high performance, data integrity, and efficient storage solutions.
- Designed and implemented high-performance NoSQL database solutions using DynamoDB, ensuring fast and reliable data access for real-time applications.
- Developed and optimized complex SQL queries to enhance data manipulation and retrieval, supporting advanced analytics and reporting.
- Implemented machine learning models using Spark MLlib for predictive analytics, supporting data-driven decision-making and enhancing business intelligence.
- Collaborated with cross-functional teams, supported DevOps practices using Jenkins and GitHub, and integrated tools like Databricks and Snowflake to optimize data workflows and ensure seamless deployment processes.
- **Environment:** Python, AWS (S3, Redshift, EMR), Apache Beam, Apache Spark, AWS Glue, Apache Kafka, Apache Airflow, Data bricks, Splunk, Jenkins, GitHub, DynamoDB, SQL.

Ericsson

January 2020 – July 2022

Data Engineer

New York, US

- Leveraged AWS S3 and Redshift for secure and scalable data storage, enabling efficient data management and analytics in the insurance sector.
- Designed and optimized data pipelines in Scala with Apache Beam and Apache Spark, ensuring robust handling of both batch and streaming data workflows.
- Implemented ETL processes with AWS Glue, Apache Kafka, and MongoDB, improving data integration efficiency by 25%.
- Utilized AWS EMR for large-scale data processing, optimizing resource usage by 35% and enhancing processing speed.
- Integrated Apache Airflow to automate complex data workflows, enhancing operational reliability and pipeline management.
- Deployed Splunk and Dynatrace for monitoring, reducing issue resolution time by 50% and improving system reliability.
- Developed Spark Structured Streaming applications for real-time data processing, improving application responsiveness and accuracy.
- Ensured compliance with financial regulations and implemented robust data security measures to maintain high data integrity and protection
- **Environment:** AWS services (S3, Redshift, EMR, Schema Conversion Tool), MongoDB, Scala, Apache Kafka, Apache Airflow, Hadoop YARN, Python, SQL, SSIS, compliance with financial regulations, data security, data quality, Shell Scripting

Wellmark

May 2019 – December 2019

Data Engineer

Iowa, US

- Extracted data from relational and NoSQL databases, including PostgreSQL and MongoDB, ensuring comprehensive data coverage for downstream analytics and reporting.
- Designed and implemented robust data cleaning pipelines in Python, addressing inconsistencies, duplicates, and missing values to enhance data reliability.
- Optimized data ingestion and transformation into AWS S3 and Snowflake, reducing query response times by 40% and improving storage efficiency.
- Contributed to the implementation of Data Mesh Architecture, enhancing decentralized data management and boosting cross-team data accessibility.
- Developed and maintained complex SQL and NoSQL databases, integrating custom ETL solutions using Java for scalability and reliability, improving analytics data usability.
- Collaborated with stakeholders across departments to align engineering efforts with business objectives, accelerating project delivery timelines by 20%.
- Built APIs with Node.js to facilitate real-time data access and streamline communication between systems
- **Environment:** PostgreSQL, MongoDB, AWS Glue, Python, Java, Node.js, S3, Snowflake, Apache Spark, AWS IAM, AWS CloudWatch, JIRA.

Prudential Financial

June 2018 – December 2018

Data Engineer

India

- Designed and implemented ETL pipelines using Apache Spark and AWS Glue, streamlining data transformation and integration, and improving overall data processing efficiency.
- Developed automated scripts in Python for data cleaning and validation, enhancing data consistency and accuracy by addressing duplicates, missing values, and formatting issues.
- Built and maintained data workflows using Apache Airflow and Kafka for batch and real-time data streaming, reducing processing delays and improving pipeline reliability.
- Managed data storage and retrieval in AWS S3 and Redshift, optimizing query performance and ensuring scalable, secure data warehousing solutions.
- Created interactive dashboards in Tableau to visualize key metrics and trends, enabling stakeholders to derive actionable insights and drive data-driven decision-making.

Environment: Python, Apache Spark, AWS S3, Redshift, Tableau, Apache Airflow, SQL, Jenkins, GitHub

Education

University of Maryland, Baltimore County

August 2022 – May 2024

Master of Science in Data Science

Maryland, US

Certifications

- Python Programming certification from Hacker Rank coding platform
- AWS Certified Solutions Architect - Associate